

Théo Alessandro Hermann

Independent robotics engineer · MIT Senseable City Lab Rio

thermann@mit.edu · thermann.ai@gmail.com · github.com/thrmnn · linkedin.com/in/theohermann-epfl · theoalessandro.com

MSc Robotics, EPFL. Four years shipping perception and autonomy systems on edge hardware. Franco-Brazilian, trained in Switzerland. French · English · Portuguese · Spanish.

First engineering hire (perception) at **Roboat** (MIT spinoff), through deployment on Amsterdam’s canals. Research fellow at **MIT Senseable City Lab Rio** — terrestrial LiDAR-based 3D reconstruction of dense informal settlements. Research assistant at EPFL VITA Lab; visiting researcher at IRI (CSIC-UPC), Barcelona.

EXPERIENCE

Research Fellow — Founding Member, Rio branch Jan 2025 – Present

MIT Senseable City Lab Rio · Rio de Janeiro, Brazil

Researcher on the urban-digital-twin line — high-fidelity 3D reconstructions of favelas from terrestrial LiDAR and aerial imagery, feeding OpenFOAM CFD simulations of airborne disease transmission. Tech: Python, Blender, GeoPandas, Google Earth Engine.

Perception Engineer — First Engineering Hire Mar 2024 – Aug 2024

Roboat (MIT Spinoff) · Amsterdam, Netherlands

Designed and shipped end-to-end perception stack for autonomous canal vessels. YOLO-based detection + multi-object tracking at <30ms on Jetson Orin. Built full MLOps pipeline with DVC over 100+ hours of maritime data. LiDAR point cloud segmentation for dynamic water filtering. Tech: PyTorch, ROS2, Docker, DVC.

Visiting Research Fellow Oct 2023 – Feb 2024

IRI (CSIC-UPC) · Barcelona, Spain

Built PyTorch fusion model combining neuromorphic event cameras with visual-inertial odometry for 6-DOF drone state estimation during aggressive maneuvers (>5g). Tech: PyTorch, ROS, C++.

Research Assistant Jun 2022 – Jun 2023

VITA Lab, EPFL · Lausanne, Switzerland

Built and open-sourced a full ROS autonomy stack for mobile robot navigation in crowded environments. Real-time detection, multi-object tracking, 3D pose estimation, and trajectory prediction on Segway Loomo. Tech: ROS, Python, C++, OpenCV.

EDUCATION

Master’s degree (M.Sc), Robotics — EPFL 2021–2024

Exchange (B.Sc 3rd year), Embedded Systems — KTH Royal Institute of Technology 2020–2021

Bachelor’s degree (B.Sc), Microengineering — EPFL 2018–2021

TECHNICAL SKILLS

Robotics	ROS/ROS2, SLAM, path planning, sensor fusion, Jetson/edge deployment
ML / AI	PyTorch, scikit-learn, reinforcement learning
Perception	LiDAR processing, point clouds, stereo vision, event cameras, 3D reconstruction
Dev	Python, C++, Git, Docker, CI/CD, DVC, MLOps
Tools	OpenFOAM CFD, Blender, GeoPandas

LANGUAGES

French Native · **English** Fluent · **Portuguese** Fluent · **Spanish** Conversational